HabitForge

Habit & Consistency Tracker

Authors	**Course**
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1. Project Description

HabitForge is a personal productivity and consistency tracking platform designed to help users build and maintain structured habits, from daily coding practice and gym workouts to reading and interview preparation.

Many students and professionals struggle with consistency due to lack of measurable, real-time feedback. HabitForge solves this by providing:

- A clean dashboard to create and manage habits with defined frequencies (daily or weekly)
- Completion logging by date to record and review progress over time
- Streak tracking (current and longest) to gamify consistency
- Weekly completion percentage analytics to identify trends and performance gaps
- Sort and filter capabilities to surface strengths and areas for improvement

The application is built with:

Backend	Database	Frontend
Node.js + Express	MongoDB (Node Driver)	React with Hooks

2. User Personas

Three primary user personas have been identified based on common productivity and self-improvement goals:

Persona 1	**Alex – Productivity-Focused CS Student** *Age: 21, Role: Junior CS Student, Goal: Land a software engineering internship* Alex is preparing for internship season and wants to maintain a daily LeetCode and coding practice habit. He knows consistency matters but often loses track of whether he practiced 3 days ago or yesterday. He needs a simple tool that shows him at a glance how many days in a row he has been coding, and flags when he is about to break his streak. **Needs:** Streak visibility, daily habit logging, simple UI, quick log entry

Persona 2	**Jordan – Fitness-Oriented Professional** *Age: 29, Role: Marketing Manager, Goal: Maintain fitness and mindfulness routine* Jordan has a busy work schedule and wants to keep gym (3x/week) and meditation (daily) habits on track. She does not want a complex app, just something that lets her log a workout and see how her week is going. She wants a weekly completion percentage to measure whether she met her targets each week. **Needs:** Weekly frequency habits, % completion stats, clean mobile-friendly layout

Persona 3	**Sam – Goal-Oriented Beginner** *Age: 24, Role: Recent Graduate, Goal: Build sustainable life routines* Sam is new to habit tracking and has struggled with apps that are too complex or overwhelming. He wants something beginner-friendly that gives him structured feedback, like a weekly summary showing how many out of 7 days he read or journaled. He responds well to positive reinforcement and wants to see improvement week over week. **Needs:** Analytics insights, week-over-week comparison, encouraging feedback, easy onboarding

3. User Stories

3.1 Habit Management & Logging (Aniket Nandi)

#	User Story	Acceptance Criteria	CRUD Operation
US-1	As a user, I want to create a new habit with a name, category, and target frequency (daily or weekly), so I can track specific goals.	Form validates name, category, and frequency fields. New habit appears in dashboard immediately.	CREATE – POST/api/habits
US-2	As a user, I want to edit a habit's name, category, or frequency, so I can update goals as my needs change.	Edit modal pre-fills existing data. Saved changes reflect immediately on dashboard.	UPDATE – PUT/api/habits/:id
US-3	As a user, I want to delete a habit and all its logs, so I can remove goals that are no longer relevant.	Confirmation prompt shown before deletion. Habit and all related logs are removed.	DELETE – DELETE/api/habits/:id
US-4	As a user, I want to log a completion for a specific habit on a specific date, so I	Date picker allows past dates. Duplicate entries for the same date are prevented	CREATE – POST/api/logs

	can record my progress accurately.		
US-5	As a user, I want to view all logged completions for a habit, so I can see my full consistency history.	Log list shows date and timestamp. Sorted by most recent first.	READ – GET/api/logs/:habitId
US-6	As a user, I want to delete a logged completion entry if I recorded incorrect data.	Delete button on each log entry. Confirmation prompt prevents accidental deletion.	DELETE – DELETE/api/logs/:id
US-7	As a user, I want to register an account so I can securely access my personal habits	Registration form validates username and password. User is logged in immediately after registering	CREATE – POST /api/auth/register

3.2 Analytics & Insights (Runze Wang)

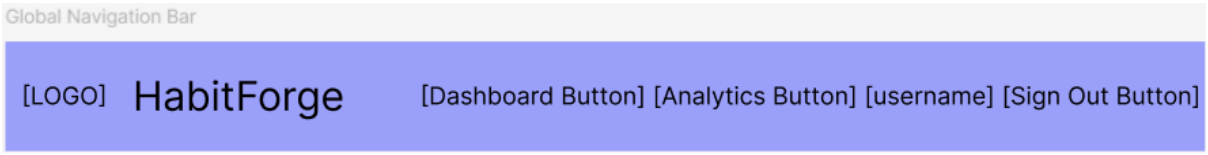
US-8	As a user, I want to see my current streak for each habit, so I can measure my ongoing consistency.	Streak calculates consecutive days with logs up to today. Resets to 0 if a day is missed for daily habits.	READ – GET/api/analytics/streak/:habitId
US-9	As a user, I want to see my longest streak for each habit, so I can evaluate improvement over time.	Longest streak displayed alongside current streak. Historical max persisted across log deletions.	READ – GET/api/analytics/streak/:habitId
US-10	As a user, I want to view my weekly completion percentage for each habit, so I can understand performance trends.	Percentage = (logs this week/target frequency) x 100. Shown as progress bar and numeric value.	READ – GET/api/analytics/weekly/:habitId
US-11	As a user, I want to sort my habits by completion rate, so I can identify strengths and weaknesses.	Sort toggle on analytics page. Default sort is by completion rate descending.	READ – GET/api/analytics/summary(with sort param)
US-12	As a user, I want to filter analytics by a custom date	Start and end date pickers. Analytics recalculate	READ – GET/api/analytics/range/:habitId?start=&end=

	range, so I can analyze specific time periods.	dynamically on date change. Validates date range.	
US-13	As a user, I want to set a weekly completion goal for each habit, so I can measure my performance against a personal target.	Goal form appears on Analytics page under each habit. User can set a target percentage between 1-100. Saved goal persists on page refresh. Only one goal allowed per habit.	CREATE – POST /api/goals READ – GET /api/goals/:habitId UPDATE – PUT /api/goals/:id DELETE – DELETE /api/goals/:id
US-14	As a user, I want to log in and out of my account so my data stays private	Login form authenticates credentials. Session persists on refresh. Logout clears session	READ/CREATE – POST /api/auth/login, POST /api/auth/logout, GET /api/auth/me

4. Design Mockups

The following mockups describe the four core screens of HabitForge. Each screen shows the layout structure, primary UI elements, and interaction zones. All screens share a persistent top navigation bar.

4.0 Global Navigation Bar



4.1 Dashboard – Habit Overview

Dashboard

[Total Habits Count] [Showing Habits Count] [+New Habit Button]

[Search Habits Bar] [Category Filter] [Frequency Filter]

[LOGO] [Habit Name] [Habit Category]
[Frequency] Daily or 3x/week
[View Logs Button] [Edit Button] [Delete Button]

[LOGO] [Habit Name] [Habit Category]
[Frequency] Daily or 3x/week
[View Logs Button] [Edit Button] [Delete Button]

4.2 Add/Edit Habit Modal

Add/Edit Habit Modal

Create New Habit

Habit Name
[Habit Name Input Field]
Category
[Category Drop Down Selector]
Frequency
[Daily Selector] [Weekly Selector]

[Cancel Button] [Create Habit Button]

4.3 Log View – Habit Completion History

Log View

[Back to Dashboard Button]

[Habit Name]

[Category] [Frequency] [Total Log Count]

LOG A COMPLETION
Date Notes (Optional)
[Date Selector] [Notes Text Field optional] [+Log Completion Button]

History

[Tick Logo] [Day, Date]
[Logged at Time] [Delete Button]
[Personal best display]

[Tick Logo] [Day, Date]
[Logged at Time] [Delete Button]
[Personal best display]

4.4 Analytics Dashboard

Analytics Dashboard

Analytics

[SUMMARY CARDS]

[LOGO] [Best Active Streak Counter] Best Active Streak	[LOGO] [Avg Weekly Percentage] Avg Weekly %	[LOGO] [Total Logs Counter] Total Logs	[LOGO] [Longest Ever Streak Counter] Longest Ever Streak

Date Range
[Selectors - This week/month/Past Year]
From To
[Date Selectors] [Apply Filter button]
Sort Habits
Sort By: [Sort Selectors - Completion% / Current Stream / ...]

[LOGO] Top Habit This Week: [Habit Name] [% completion]

[LOGO] [Habit Name] [Category] [Current/best/total log mini cards]
Weekly [Progress bar..... & %]
Overall [Progress bar..... & %]

[LOGO] Weekly Goal +[Set Goal Button] or [Target Display] [Edit & Delete Buttons]

4.5 Login Screen

Login Screen

[LOGO] HabitForge

Sign in to your account

Username

[Username Text Field]

Password

[Password Text Field]

[Sign In Button]

Don't have an account? [Register btn]

4.6 Register Screen

Register Screen

[LOGO] HabitForge

Create your account

Username

[Username Text Field]

Password

[Password Text Field]

Confirm Password

[Password Text Field]

[Create account button]

Already have an account? [SignIn btn]